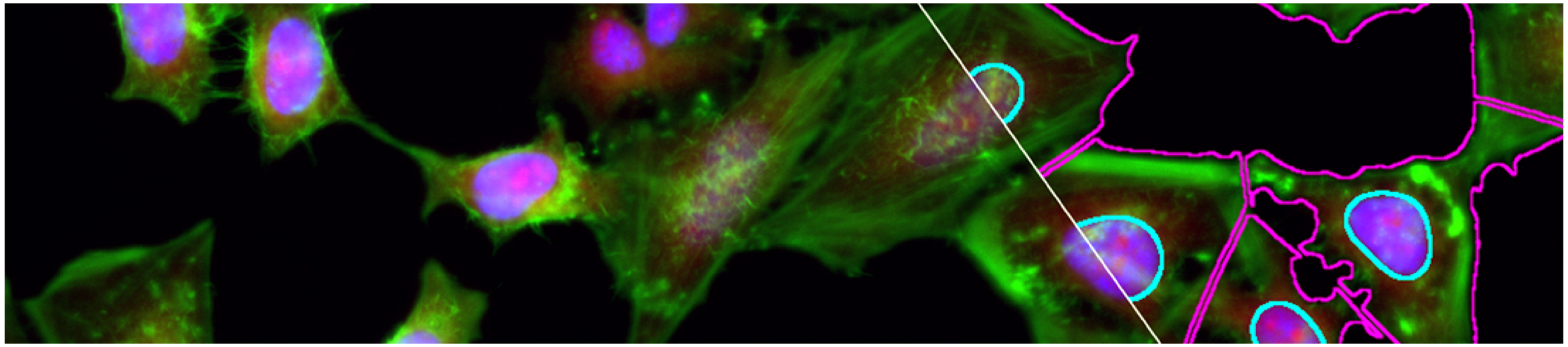
7.1.1. Description & Outline



Week Overview

- Section 1: Introduction
- Section 2: Color and Color Models
- Section 3: Projections
- Section 4: Reslicing
- Section 5: Conclusion

Description

In the past weeks, we built up momentum until we were able to understand and produce a complete image analysis workflow through understanding digital images, their dimensions, what kind of operations we can do on them, filtering and segmentation. With the knowledge you have accumulated, you should be well armed to construct simple image analysis workflows, and to understand and adapt more complex ones to your needs.

This final week takes a break to show you a series of essential tools and techniques to extract more information from images and ease segmentation and/or quantification. We start by revisiting color images and talking about color spaces and color deconvolution, to ease extraction of meaningful colors.

We then cover a concept called 'dimensionality reduction', which deals with how data can be projected to keep as much information as possible while reducing the complexity of multidimensional data. This can be particularly useful when presenting data for publication purposes.

Finally, we cover reslicing of multidimensional images, in both space and time, which can allow for more straightforward quantification. One such method you will discover are kymographs.

By the end of the week, you will be able to list at least 3 different color spaces and list their pros and cons as well as define the idea behind color deconvolution. You will be able to name 2 ways of reslicing image data and apply it in order to extract quantitative data. Finally you will be able to define image projections, list and use at least 3 different projection methods.

Resources

Week 7 Images: (ZIP, 123MB) The image data required for Week 7 exercises